Spectral decomposition for efficient smoothing-parameter selection in penalized least squares: an empirical study of complete-criterion computation

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Abstract

Selecting the smoothing parameter of a finite-difference penalized least-squares estimator requires evaluation of a statistical selection criterion, not merely solution of a penalized linear system. We examine the computational advantages and numerical limitations of reusing a singular-value decomposition (SVD) across candidate parameters, compared with a structured banded-Cholesky implementation. The study covers 29 prespecified configurations, 97 random-seed realizations per configuration, multiple selection criteria and one- versus multiple-candidate workloads. Valid execution times are matched by realization, workload and execution phase, with uncertainty estimated by seed-cluster bootstrap. For an order-three conditional cross-validation problem with $\boldsymbol{n = 256}$, the banded/SVD complete cold-time ratio is 0.698 (95% interval 0.671–0.724) for a single candidate but 4.846 (4.567–5.125) for a 100-candidate grid. In nine configurations, the banded complete-grid implementation fails its post-solve optimality check for all recorded grid evaluations while the spectral pipeline completes them; both methods exhibit smoothness-inversion failures in larger fourth-order configurations. Dense spectral storage nevertheless scales quadratically with series length. The results support spectral decomposition as the preferred implementation for repeated complete-criterion searches when memory permits, while retaining banded methods for isolated evaluations and memory-constrained problems.

Keywords: penalized least squares, smoothing-parameter selection, singular-value decomposition, Whittaker–Henderson smoothing, cross-validation, numerical reliability, computational performance

1 Introduction

Penalized least-squares smoothing is a widely used approach to separating low-frequency variation from noise in an observed series. Since the work of Whittaker Whittaker (1923), finite-difference penalties have offered a direct connection between trend regularity and linear smoothing. Subsequent developments include structured implementations of Whittaker–Henderson smoothing Weinert (2007, 2012); Biessy (2026) and the interpretation of smoothing through the percentage-of-smoothness parameter introduced by Guerrero Guerrero (2007, 2008). Selection of the penalty by cross-validation, information criteria or marginal likelihood is well established Craven and Wahba (1979); Cortés-Toto et al. (2017). Nevertheless, obtaining a fitted trend and evaluating the *complete selection criterion* are computationally different tasks.

A banded Cholesky factorization exploits the sparsity of finite-difference penalties and is economical for an individual smoothing parameter. In contrast, a spectral factorization requires an appreciable initial calculation but exposes all candidate penalties through diagonal transformations in a common basis. Consequently, the apparent superiority of a solver in a single fit need not persist across a grid of smoothing parameters. Moreover, numerical acceptance of the complete criterion depends not only on the matrix factorization but also on leverage calculations, endpoint treatment and the inversion between penalty and smoothness coordinates.

We investigate the following practical question: *When is a spectral implementation preferable for selecting a smoothing parameter by a complete statistical criterion?* We compare measured end-to-end costs of spectral singular-value decomposition (SVD), banded Cholesky and available reference implementations under paired computational workloads. Unlike a fit-only benchmark, the comparisons charge setup, coordinate conversion, criterion evaluation and diagnostics to the method that incurs them. Unsuccessful or incomplete searches remain observable outcomes rather than being treated as fast computations. This permits a recommendation based jointly on execution time, observed reliability and storage requirements.

Our main conclusion is that spectral decomposition is an effective *default choice for repeated, complete-criterion searches at feasible matrix sizes*. That recommendation is empirical and conditional: structured banded computation remains competitive for isolated candidate evaluations and retains important storage advantages for long series. We do not propose a new PLS estimator, nor do we interpret computation time as predictive accuracy.

2 Penalized least squares and spectral evaluation

2.1 Estimator and smoothing coordinate

Let $y = (y_1, \ldots, y_n)^\top \in \mathbb{R}^n$ denote equally spaced observations and let $D_d \in \mathbb{R}^{(n-d)\times n}$ be the order-d forward-difference matrix, with $1 \le d < n$. For a fixed reference level $\mu \in \mathbb{R}$, the trend estimate is

$$\widehat{\tau}_{\lambda,\mu} = \underset{\tau\in\mathbb{R}^n}{\arg\min} \left\{ \|y - \tau\|_2^2 + \lambda \|D_d\tau - \mu\mathbf{1}\|_2^2 \right\}, \qquad \lambda \ge 0. \tag{1}$$

Writing $Q = D_d^\top D_d$ and choosing any p_μ satisfying $D_d p_\mu = \mu \mathbf{1}$, the normal equations yield

$$\widehat{\tau}_{\lambda,\mu} = p_\mu + H_\lambda(y - p_\mu), \qquad H_\lambda = (I + \lambda Q)^{-1}. \tag{2}$$

For finite $\lambda \geq 0$, $I + \lambda Q$ is symmetric positive definite. Because $\operatorname{rank}(D_d) = n - d$, the penalty matrix has precisely d zero eigenvalues. These correspond to discrete polynomials of degree at most $d - 1$.

Let $\delta_1, \ldots, \delta_{n-d} > 0$ be the positive eigenvalues of Q. The normalized smoothness parameter

$$S(\lambda) = \frac{n - \operatorname{tr}(H_\lambda)}{n - d} = \frac{1}{n - d} \sum_{j=1}^{n-d} \frac{\lambda \delta_j}{1 + \lambda \delta_j} \tag{3}$$

satisfies $S(0) = 0$ and $S(\infty) = 1$, and is strictly increasing for positive finite λ. Thus S supplies a bounded coordinate for the original estimator, without changing the underlying objective function. The endpoint $S = 1$ is handled as the constrained polynomial limit rather than by substituting an arbitrarily large finite penalty.

2.2 One decomposition, multiple smoothing parameters

Consider the singular-value decomposition $D_d = U \Sigma V_+^\top$, completed to an orthogonal basis $V = [V_+, V_0]$. Its nonzero squared singular values are the δ_j. With $a = V_+^\top (y - p_\mu)$ and $w_j(\lambda) = \lambda \delta_j / (1 + \lambda \delta_j)$, the fitted trend and residual can be represented as

$$\widehat{\tau}_{\lambda,\mu} = y - V_+ \operatorname{diag}\{w_j(\lambda)\} a, \tag{4}$$

$$r_\lambda = y - \widehat{\tau}_{\lambda,\mu} = V_+ \operatorname{diag}\{w_j(\lambda)\} a. \tag{5}$$

After the basis and series-specific coordinates have been computed, each candidate modifies only the scalar attenuation factors. The effective degrees of freedom and residual sum of squares follow from

$$\operatorname{edf}(\lambda) = d + \sum_{j=1}^{n-d} \{1 - w_j(\lambda)\}, \tag{6}$$

$$\operatorname{RSS}(\lambda) = \sum_{j=1}^{n-d} w_j(\lambda)^2 a_j^2. \tag{7}$$

In particular, a criterion requiring only RSS and edf can exploit the stored spectral coordinates without repeatedly factorizing a penalized normal matrix. This amortization is the central reason to consider SVD for a grid search.

2.3 Complete selection criteria

For a fixed numerical reference mean, ordinary leave-one-out cross-validation (CV) and generalized cross-validation (GCV) have the conventional forms

$$\operatorname{CV}(\lambda) = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{r_{\lambda,i}}{1 - h_{ii}(\lambda)} \right)^2, \tag{8}$$

$$\mathrm{GCV}(\lambda) = \frac{\mathrm{RSS}(\lambda)/n}{\{1 - \mathrm{edf}(\lambda)/n\}^2}, \tag{9}$$

where $h_{ii}(\lambda)$ is a diagonal entry of H_λ. The spectral basis also gives the stable leverage-deficit representation

$$1 - h_{ii}(\lambda) = \sum_{j=1}^{n-d} v_{ij}^2 w_j(\lambda). \tag{10}$$

Unlike GCV, ordinary CV requires observationwise residuals and leverages; consequently, the candidate-evaluation cost differs by criterion even after spectral setup. We additionally consider corrected Akaike information criterion (AICc), Bayesian information criterion (BIC), and profiled Gaussian marginal likelihood. These use the established definitions implemented consistently across the methods being compared, including their prescribed undefined points and endpoint limits. A criterion value of infinity is not interpreted as an executable failure when the criterion is mathematically undefined.

A reference mean estimated once as $\mu = \overline{D_d y}$ is used in some computational settings. In those cases the selection criteria are *conditional* on the full-sample plug-in value: the CV score does not re-estimate μ within each leave-one-out fold. This distinction is preserved throughout the analysis.

2.4 Banded evaluation and numerical admissibility

The banded method factorizes $A_\lambda = I + \lambda D_d^\top D_d$ for individual penalties. A selected-inverse recursion supplies the diagonal leverage values and effective degrees of freedom needed by the complete criteria, without forming a dense inverse. For fixed low difference order, the banded factor and selected inverse have storage proportional to nd, whereas the full spectral basis requires storage proportional to n^2. An explicit dense-inverse method and an available sparse-factorization/structured-diagnostics hybrid serve only as additional comparison procedures; the latter is not treated as a sparse-only score evaluator.

A numerical completion requirement is especially important for the banded procedure. After factorization and solution of the normal equations, the implementation checks

$$\eta_{\mathrm{KKT}} = \frac{\|(y - \widehat{\tau}) - \lambda D_d^\top (D_d \widehat{\tau} - \mu \mathbf{1})\|_2}{\max\{1, \|y\|_2, \|y + \lambda \mu D_d^\top \mathbf{1}\|_2\}}. \tag{11}$$

Its finite-precision acceptance threshold is $\eta_{\mathrm{KKT}} \leq 10^{-7}$. Rejection occurs *after* the banded factorization and solve. A failed check therefore need not imply that Cholesky factorization itself was impossible. The spectral implementation does not apply precisely the same banded-specific acceptance gate; its observed completion rate must not be confused with an independent proof of greater forward accuracy.

3 Computational study

3.1 Experimental factors and execution environment

The study evaluates 29 prespecified computational configurations and 97 independent random-seed realizations per configuration, giving 2813 scheduled case–seed jobs. Configurations vary series length n, difference order d, criterion, reference mean, signal amplitude, and grid density M. Four workload types are considered: one candidate for one series, M candidates for one series, one candidate for R series, and M candidates for R series. The tested grid is $S_j = j/M$ for $j = 1, \ldots, M$, including its analytic endpoint. Selected configurations use $M = 100$, 1000, or 10000; the examined sample lengths include $n = 64$, 128, 256, 512, 1024, 2048, and 4096. Comparisons use identical simulated series and evaluation coordinates when both implementations are admissible.

The computations were performed on a Windows x64 multicore workstation, with one BLAS thread per job. The execution schedule comprised phases allowing at most 24, 1, or 12 concurrent jobs, with 2231, 52, and 530 jobs, respectively. These limits do not measure instantaneous CPU occupancy. We therefore retain the execution phase as a matching variable and avoid interpreting multi-job wall-clock times as contention-free single-process latency. Within a case–seed, method ordering was rotated and repeated where prescribed.

Two coordinate-evaluation policies were implemented: directly inverting S for each requested candidate and converting the full S grid to λ once for reuse. Both policies target the *same* smoothing values. The reported complete cold time includes method initialization, smoothness-to-penalty conversion and all candidate criterion evaluations; recorded warm accounting excludes the one-time initialization term but does not represent a separately launched process.

3.2 Complete-search validity and paired comparisons

Every scheduled job was reconciled with its saved numerical and timing outcomes. A timing row is admitted only when the process terminates successfully, the selected score and elapsed time are finite, the recorded status is successful, and all MR prescribed evaluations are present. Missing files, rejected numerical checks, inaccurate smoothness inversions and resource-limit exclusions remain in the denominator of descriptive reliability summaries. Successful evaluation of only one grid point is never substituted for completion of the full grid.

For paired comparisons, methods must additionally agree on selected smoothness within 10^{-10} and, for directly comparable non-likelihood scores, on the selected criterion value within relative tolerance 10^{-7}. These criteria assess agreement between available double-precision implementations, not uniform high-precision accuracy throughout the parameter grid. The implementation and machine configuration, execution phase, realization, coordinate policy, workload and repeat are matched.

Let T_{B} and T_{S} denote positive complete cold times for banded Cholesky and spectral SVD. Our primary contrast is

$$\rho = \frac{T_{\mathrm{B}}}{T_{\mathrm{S}}}; \qquad \rho > 1 \iff \text{SVD is faster for that paired workload.} \tag{12}$$

Log ratios are averaged within each independent seed cluster before constructing the geometric mean. Two-sided percentile 95% confidence intervals use 700 bootstrap resamples of the independent seeds, retaining the correlated within-seed repetitions together. Confidence intervals thus characterize run-to-run variation for the measured conditions, not performance uncertainty across hardware platforms. Conditioning on numerically successful pairs can introduce survivor bias, which is addressed by reporting reliability separately from speed.

4 Empirical results

4.1 Completion rates and admissible observations

Of 2813 scheduled computational jobs, 57 terminated without a successful numerical record. The resulting dataset comprises 92,679 measured or explicitly classified method-level outcomes. A total of 76,220 (82.24%) satisfy the predefined completeness and status conditions. The remaining 16,459 comprise 11,208 numerical-rejection records, 4224 additional smoothness-inversion failure records, 970 computational-budget exclusions, and 57 records representing unavailable job inputs. The classification retains unsuccessful searches; omitting them would bias subsequent comparisons toward implementations with a higher apparent completion rate.

Figure 1 summarizes status-admissible proportions for the evaluated criteria and implementation families. Because the methods were not all attempted in the same parameter configurations, aggregate percentages should be interpreted descriptively rather than as probabilities from a randomized, identically distributed reliability trial.

4.2 Complete-grid running time

Table 1 and Fig. 2 compare complete 100-candidate CV and GCV searches for one series. Each contrast uses the same observations, numerical criterion, smoothing candidates and execution phase. The reported ratio $\rho = T_{\mathrm{B}}/T_{\mathrm{S}}$ exceeds one for all eight representative parameter configurations, indicating an advantage for spectral SVD on these measured grids. The strongest among these representative ratios exceed five, whereas the smaller first-order GCV example is approximately 2.33. These findings concern *complete selection*, including setup and diagnostics, not repeated trend fitting alone.

The effect of candidate count is particularly clear for conditional CV at $n = 256$ and $d = 3$. The banded/SVD time ratio is 0.698 (95% interval 0.671–0.724) for one candidate and one series, but 4.846 (4.567–5.125) for 100 candidates. For 100 candidates evaluated on three series, the corresponding ratio is 3.655 (3.450–3.860). The reversal is consistent with a sizable spectral setup cost that is recovered through repeated criterion evaluation. It does not imply a universal crossover at $M = 100$; both the

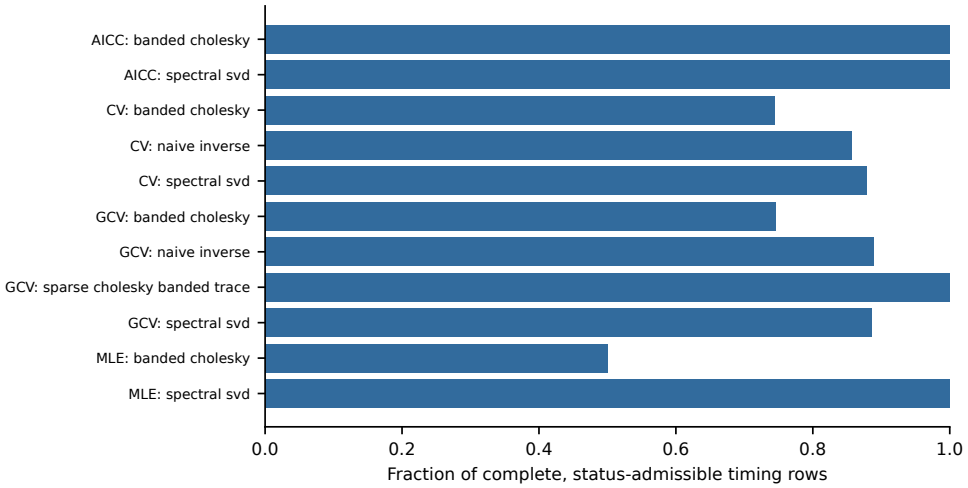


Fig. 1 Proportion of complete, status-admissible method-level observations by selection criterion and implementation. Each denominator includes recorded incomplete searches and explicit exclusions for the associated method. Differences in the evaluated configuration sets preclude interpreting aggregate percentages as matched failure probabilities.

Table 1 Matched complete-criterion cold-time ratios for 100 smoothing candidates and one series. The reference mean is fixed at zero ($\mu = 0$) or computed once from the observed differences ($\mu = \overline{D_d y}$). All comparisons use the same execution phase and smoothing-coordinate policy. Intervals are 95% percentile bootstrap intervals over independent random-seed clusters. Ratios exceeding one favor spectral SVD.

Criterion	n	d	Reference μ	Seeds	ρ	95% CI
CV	64	1	0	96	2.34	[2.29, 2.39]
CV	64	3	0	97	4.03	[3.95, 4.11]
CV	128	3	0	97	5.40	[5.24, 5.55]
CV	128	3	$\overline{D_d y}$	97	5.56	[5.46, 5.65]
CV	256	3	$\overline{D_d y}$	97	4.85	[4.57, 5.12]
GCV	64	1	0	96	2.33	[2.28, 2.37]
GCV	128	3	0	97	4.99	[4.87, 5.10]
GCV	128	3	$\overline{D_d y}$	97	5.12	[5.01, 5.23]

location and even the existence of a crossover depend on the criterion, series length, order and implementation.

For fixed-mean first-order GCV at $n = 64$, the analogous ratio is 1.001 (0.982–1.018) for one candidate and 2.328 (2.282–2.374) for the 100-candidate grid. Thus a negligible single-candidate performance difference can coexist with a substantial grid-search advantage.

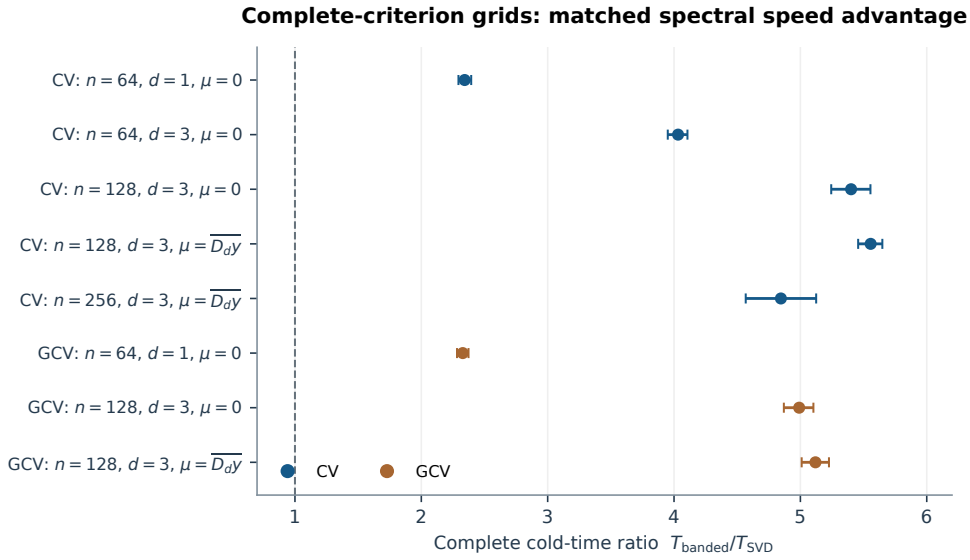


Fig. 2 Geometric-mean banded-Cholesky/SVD complete cold-time ratios for eight numerically admissible CV and GCV searches over 100 smoothness candidates. Horizontal intervals are 95% seed-cluster bootstrap confidence intervals. All plotted comparisons were matched within the same execution phase; selected contrasts have 96 or 97 independent seed realizations. The dashed vertical reference denotes equal execution time.

4.3 Reuse across smoothing coordinates and series

The candidate representation also influences total cost. Two computational policies were compared: repeated conversion of smoothness values S to the penalty λ, and a single conversion shared across the search. The candidate set is identical under both policies, so the contrast concerns computational reuse rather than a different optimization problem.

For spectral CV with $n = 256$, $d = 3$, 100 candidates and three series, the geometric complete cold-time ratio of once-mapped penalties to repeated direct-S conversion is 0.580 (95% interval 0.570–0.590). In this setting, precomputing the candidate transformation reduces total execution time. Figure 3 displays corresponding descriptive comparisons across other numerically admissible multi-series grid configurations.

Figure 4 illustrates how a fixed first-order GCV configuration responds to four combinations of candidate and series count. The comparisons show why initialization, single-candidate scoring, multiple-series reuse and full-grid selection must be distinguished when assessing an implementation. In particular, a procedure optimized only for one fitted trend need not minimize the cost of selecting a smoothing parameter.

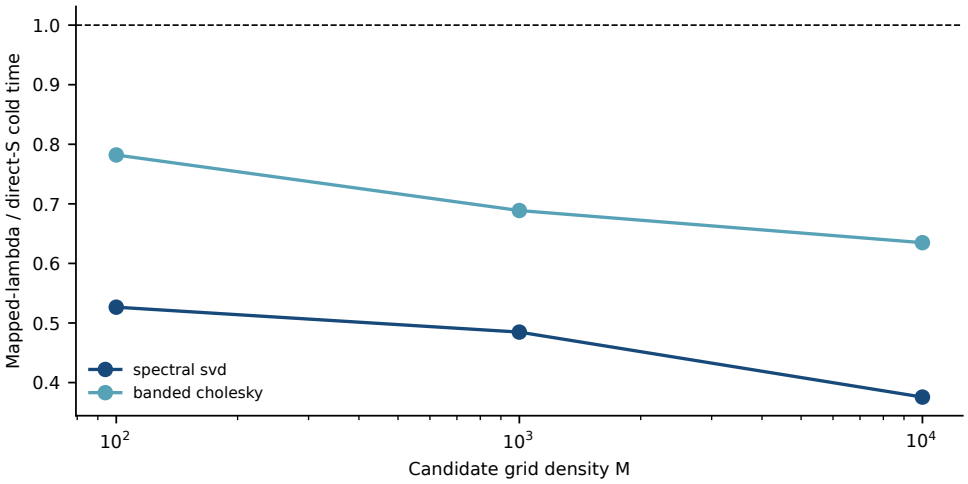


Fig. 3 Matched cost of once-mapped penalties relative to direct smoothness-coordinate conversion in multiple-series grid workloads. Plotted values summarize geometric time ratios within admissible configurations from the same execution phase. Values below one favor a shared penalty mapping; the aggregates are descriptive rather than universal mappings from grid size to execution time.

4.4 Numerical reliability and its consequences

The aggregate status proportions in the 24-worker execution phase are 10,848/13,176 (82.3%) for banded CV and 13,176/13,176 (100%) for spectral CV. The corresponding GCV counts are 13,176/16,280 (80.9%) and 15,892/16,086 (98.8%), respectively. Their denominators differ because method eligibility and the evaluated configurations differ; they should not be used as an unconditional estimate of solver accuracy. The comparison nevertheless motivates a closer examination of the failure mechanisms, reported next.

4.5 Post-solve optimality failures in banded computation

The banded-Cholesky implementation reported 9096 method-level timing observations rejected by the relative optimality-residual check in Eq. (11). The recorded residuals range from approximately 1.00×10^{-7} to 1.06×10^{-6}, compared with the prescribed tolerance 10^{-7}. Since the check is performed after factorization and solution, these observations do not establish the breakdown of Cholesky factorization itself. They identify loss of admissibility in the *complete banded criterion pipeline* under its implemented numerical diagnostic.

A particularly informative contrast occurs in nine parameter configurations. For each configuration, 97 independent realizations were considered. Both methods completed every recorded single-candidate evaluation, but the banded implementation rejected every complete-grid evaluation, whereas the spectral implementation completed the corresponding grids. The configurations comprise CV and GCV with $(n, d, M) = (128, 3, 1000)$, $(256, 4, 100)$ and $(256, 3, 1000)$; GCV with $(64, 3, 10000)$;

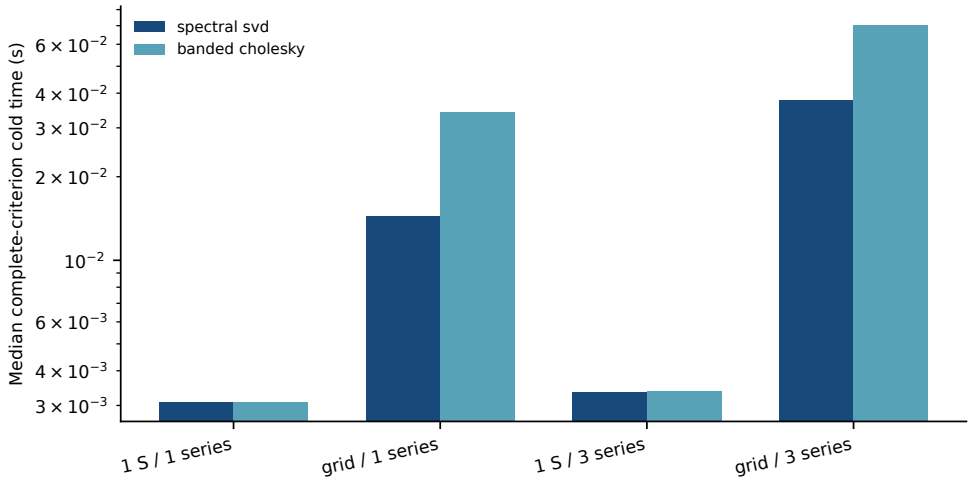


Fig. 4 Median complete-criterion cold time for the four workload types in a first-order GCV example ($n = 64$, 100-candidate grid). The reported time includes initialization, coordinate mapping and all evaluated criterion values. The within-seed workload observations are correlated.

and profiled Gaussian marginal likelihood with $(256, 3, 100)$ and $(512, 3, 100)$. Figure 5 displays these results.

The nine configurations account for 6984 banded-rejection timing records; another 2112 such records arise from larger fourth-order configurations. These are repeated method-level observations within 97 independent seed clusters per configuration, rather than thousands of independent experimental realizations. The saved status information does not identify the exact smoothing-grid point that first produced each rejection, and hence no candidate-specific instability boundary is inferred.

The apparent spectral advantage must be interpreted as *observed computational completion*, not a controlled comparison of identical backward-error tests. The spectral scorer does not impose the same implementation-specific acceptance criterion as the banded scorer. Independent numerical reference calculations would be required to establish a strict relative-accuracy ranking at each penalty.

4.6 Smoothness inversion remains a common numerical limitation

An additional failure mechanism affects both implementations at larger difference order and series length. In the fourth-order CV/GCV configurations with $n = 1024$, 2048 and 4096, the numerical inversion of the normalized smoothness mapping sometimes returned values outside its prescribed tolerance. Across the evaluated configurations, 6336 timing records explicitly describe inaccurate smoothness inversion: 2112 associated with the banded method and 4224 with SVD. The distribution of observed outcomes in the fourth-order settings is given in Fig. 6.

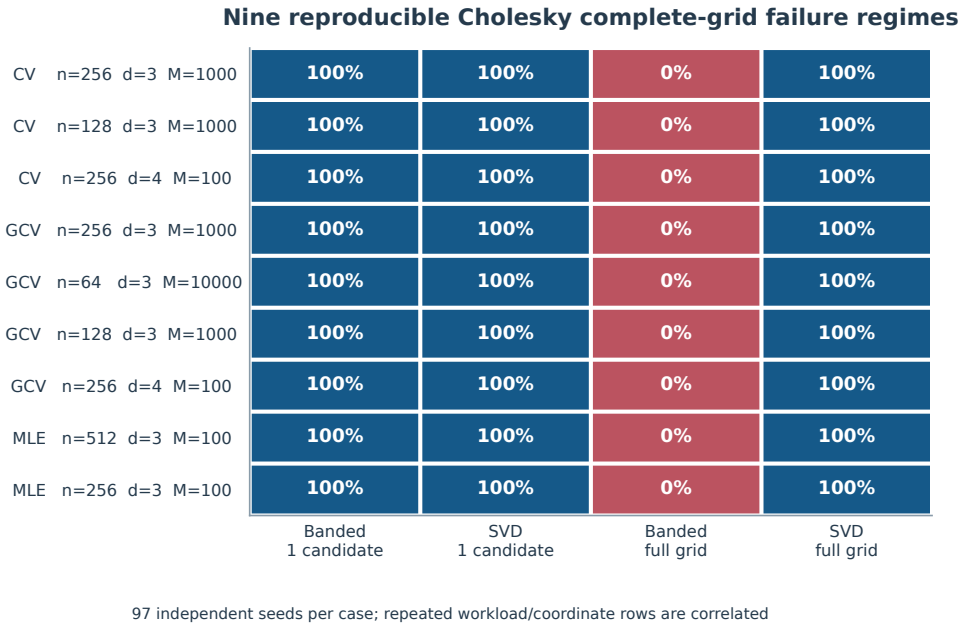


Fig. 5 Observed completion fractions for nine parameter configurations contrasting a single candidate with a full smoothing grid. Each configuration contains 97 independent random-seed realizations. The banded grid fails the post-solve optimality check for all recorded grid observations, whereas the spectral pipeline completes the corresponding grids. Repeated measurements and candidate policies within each realization are not independent trials.

This error concerns the transformation from the bounded coordinate S to the penalty λ, rather than evidence that a spectral decomposition failed. Crucially, a successfully completed one-candidate evaluation cannot be used as a substitute for an incomplete multi-candidate search. Accordingly, the large-order configurations with shared inversion difficulties do not supply an empirical winner for the full criterion grid.

4.7 The single-candidate versus grid-search crossover

The principal runtime trade-off becomes visible when paired, numerically admissible observations are compared across candidate counts. Figure 7 shows three examples evaluated under identical matching rules. In conditional CV with $n = 256$, $d = 3$, the geometric banded/SVD cold-time ratio changes from 0.698 (95% interval 0.671–0.724) for one candidate to 4.846 (4.567–5.125) for 100 candidates. In fixed-mean GCV with $n = 64$, $d = 1$, the corresponding ratios are 1.001 (0.982–1.018) and 2.328 (2.282–2.374). For fixed-mean CV with $n = 128$, $d = 3$, they are 1.069 (1.052–1.087) and 5.401 (5.244–5.555).

These results support the computational interpretation of Eqs. (4)–(7): spectral setup is paid once, while the attenuation coefficients and the statistics needed for

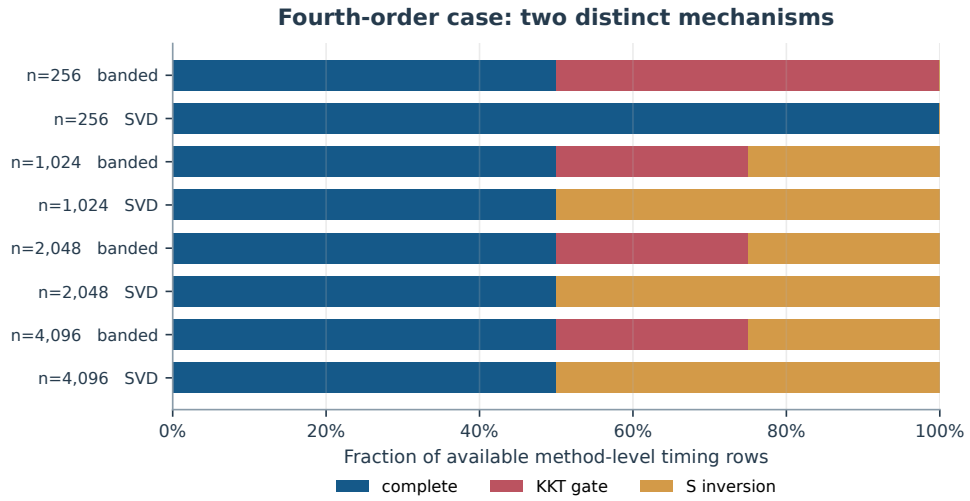


Fig. 6 Status composition of the fourth-order computations at increasing series lengths. The displayed denominator is the number of available method-level records. Banded post-solve optimality failures and smoothness-coordinate inversion failures are distinguished; the latter affect both implementations. Excluded or unavailable jobs are retained separately in the overall accounting.

subsequent criterion evaluations are reused. The crossover is observed in complete scoring workloads; it should not be conflated with the cost of fitting a trend for a prescribed penalty.

4.8 Memory requirements and computational feasibility

The principal disadvantage of the spectral approach is its dense storage requirement. For the evaluated implementation, the conservative memory admission allowance corresponds to five $n \times n$ double-precision arrays, or $40n^2$ bytes. An explicit dense-inverse reference is assigned seven such arrays, or $56n^2$ bytes. In contrast, a banded Cholesky factor and a banded selected-inverse buffer together require approximately $16n(d+1)$ bytes, excluding ancillary data structures. Figure 8 illustrates these storage relations for $d = 4$.

For example, the spectral admission allowance is 640 MiB at $n = 4096$. At $n = 8192$ it rises to 2.5 GiB, exceeding the 2 GiB dense allocation limit imposed in the study. The latter sample length was not evaluated in the present comparison. The curves therefore represent *implementation-specific memory allowances and illustrative buffer sizes*, not measurements of peak resident memory or empirical allocation failures.

There were 970 resource-budget exclusions in the recorded timing outcomes: 776 for the explicit-inverse reference and 194 for SVD. These were induced by prescribed computational-work thresholds, not observed out-of-memory errors. Thus the memory analysis establishes a concrete feasibility trade-off without attributing numerical failures to unsupported memory measurements.

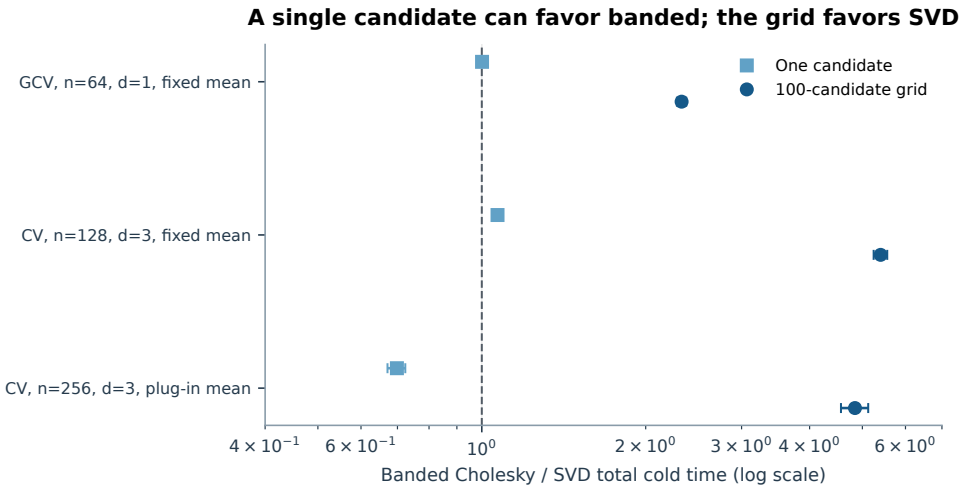


Fig. 7 Matched banded-Cholesky/SVD complete cold-time ratios for one smoothing candidate and for 100 candidates. Horizontal bars represent 95% seed-cluster bootstrap intervals. Values above one favor spectral SVD. Only complete and numerically admissible method pairs contribute to the displayed estimates.

5 Discussion

The empirical evidence supports a clear implementation preference, subject to mathematical and computational feasibility: *reuse of a spectral decomposition is the recommended default for finite-difference PLS smoothing-parameter selection over a moderate-to-large candidate grid*. In contrast to a sequence of penalized normal-equation factorizations, spectral preprocessing represents the complete search in one orthogonal basis. It amortizes setup, supplies degrees of freedom and residual sums of squares directly, and provides a common representation for multiple selection criteria. The advantage is pronounced in the matched CV/GCV grid comparisons and is compatible with the observed numerical-completion differences.

The recommendation should not be extended beyond the observed evidence. First, the banded method is often preferable when evaluating only one penalty. Second, the spectral basis and its work arrays require quadratic memory, which can dominate the decision for long series even when SVD is attractive at moderate n. Third, numerical acceptance is not interchangeable with certified forward accuracy: the two implementations apply different diagnostic checks, and both encounter smoothness-inversion difficulties in the larger fourth-order cases. Fourth, the recorded multi-process execution phases limit the transportability of wall-clock ratios to idle workstations or different parallel workloads.

These limitations suggest the following practical selection principle. If the task is a multi-candidate *complete-criterion* search and the dense spectral workspace satisfies a conservative memory budget, prepare the spectral decomposition once, reuse series-specific projections and precompute the candidate penalties when appropriate.

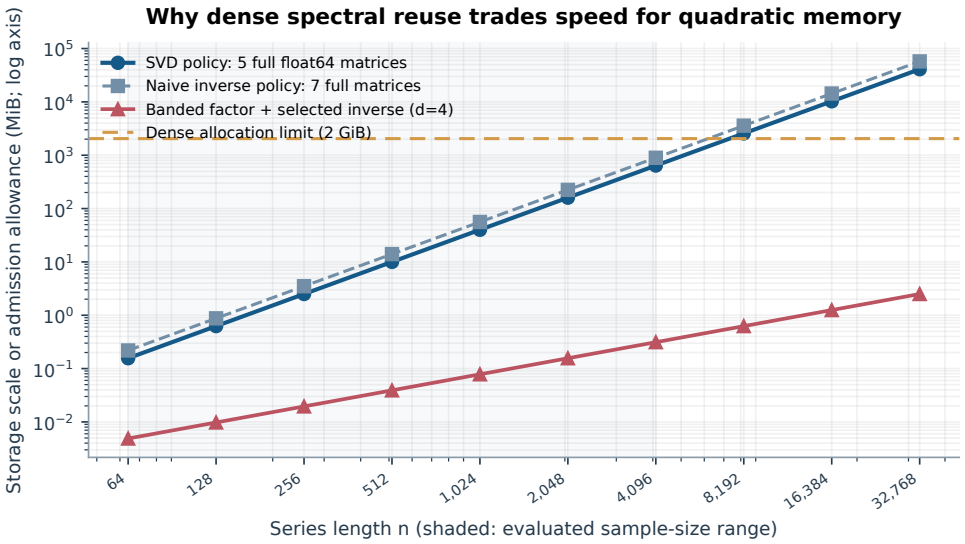


Fig. 8 Quadratic dense memory allowances for spectral SVD and an explicit-inverse reference compared with the storage of two order-four banded working arrays. The horizontal line denotes the computational study's 2 GiB dense allocation limit. These are analytical, implementation-specific storage calculations rather than measured peak memory; only the shaded sample-size range was included in the study.

If only one candidate is needed, or if dense allocation would be prohibitive, a structured banded method is a reasonable alternative subject to numerical residual checks. When a requested smoothness value cannot be inverted accurately, neither method should silently substitute an approximate grid winner: the affected comparison must be marked invalid or handled by a separately validated numerical strategy.

A further limitation is the restriction to synthetic data and one computational environment. Seed-cluster intervals quantify variability under the observed experimental protocol, but not variability across operating systems, processors, library versions or CPU utilization. The validity masks can induce survivor bias in timing ratios conditional on completed searches. The study does not establish improvements in trend-estimation risk or forecasting accuracy, and it does not constitute an independent high-precision verification of every evaluated penalty. Future work should isolate the source of the large-order smoothness-inversion failures, independently certify residual and criterion accuracy, and compare implementations under controlled processor occupancy.

6 Conclusions

This study establishes that the preferred computational method for penalized least-squares trend smoothing depends on the *parameter-selection workload*, not merely the cost of one linear solve. Across valid matched 100-candidate CV and GCV searches,

spectral SVD consistently outperformed banded Cholesky in the representative tested configurations. A measured conditional-CV setting changed from favoring banded Cholesky for one candidate to favoring spectral SVD by a factor of approximately 4.85 for the complete grid. The numerical-status analysis further identified repeated banded post-solve failures on several complete grids, while retaining documented failure modes of the spectral pipeline.

We therefore recommend *spectral decomposition as the primary implementation for complete, repeated smoothing-parameter selection whenever its dense storage is affordable*. Banded methods remain valuable for isolated evaluations and long series, and numerical diagnostics remain mandatory for either implementation. The contribution is a measured, criterion-aware recommendation for computational practice rather than a new trend estimator or an unconditional claim of spectral superiority.

7 Code and data availability

The source implementations, analysis programs, selection-status records, derived timing data and reproducible figures are available in the public project repository at https://github.com/heri-espino/trendestimation. The underlying raw execution records are retained by the author but are not publicly deposited; their availability is subject to author approval. The publicly archived derived observations preserve incomplete and numerically inadmissible searches, and include the pairing and bootstrap metadata required to reproduce the tabulated comparisons.

Declarations

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Competing interests. No competing interests have been reported.
Ethics approval and consent. The study consists of synthetic numerical experiments and does not involve human participants, identifiable personal data or animal subjects.
Data and code availability. Analysis code, derived benchmark results, numerical-status tables and figures are available at https://github.com/heri-espino/trendestimation. Underlying raw timing files remain preserved by the author but are not part of the public repository; access is subject to author approval.

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